BreakfastWise PRD

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1. Product summary

BreakfastWise tells a hotel's head chef, at about 9 PM, how many guests will eat breakfast tomorrow and how much of 5 key dishes to prepare, and shows recent waste and stock-outs. It uses only occupancy and past breakfast records, with simple arithmetic a judge can check by hand.

Primary user: Chef Murugan, head chef of an independent 100-room business hotel in Trichy (see the MRD). **Secondary user:** the F&B manager reviewing results later.

MRD needs this PRD serves

ID	Need (from the MRD)
N1	Know each evening how many guests will eat breakfast tomorrow
N2	Know how much of each dish to prepare, without over- or under-preparing
N3	See where food was wasted or ran out, so mistakes are not repeated
N4	Get a consistent, checkable estimate instead of a guess that varies by person and day

Data note: The demo uses 60 days of clearly labelled synthetic data. Suggestions are described as suggested quantities, not validated demand predictions.

2. Features and priorities

Three Must-Haves carry the chef's 9 PM workflow: forecast the guests, plan the dishes, review waste. Each passed the demo test: if it fails, the demo loses its core story.

ID	Feature (capability)	MoSCoW	MRD need
F1	Chef can enter tomorrow's expected guests and see predicted breakfast guests, with the calculation explained	Must	N1, N4

ID	Feature (capability)	MoSCoW	MRD need
F2	Chef can see suggested quantities for 5 dishes (idli, dosa, pongal, poori, sambar) with a default 10% safety buffer shown	Must	N2
F3	Chef can see recent prepared, consumed and discarded quantities per dish, with stock-out days flagged	Must	N3
F4	Chef can adjust the safety buffer	Should	N2
F5	F&B manager can compare forecast with actual breakfast guests for past days	Should	N4
F6	Chef can enter a day's actual figures through a form	Should	N3
F7	Manager can view trend charts of attendance and waste	Could	N3
F8	Machine-learning forecast model	Won't	N1
F9	Automatic import from hotel management systems (PMS)	Won't	N1

Why each Must-Have

- **F1:** Without the guest forecast there is no product; F2 depends on it.
- **F2:** The chef's real decision is quantity, so a guest number alone only half solves waste.
- **F3:** Waste and shortages are the problem statement; without F3 the demo cannot show either.

Historical data is preloaded from a CSV as a build task inside F1, so a broken entry form (F6) cannot block the demo.

3. Forecast logic

The forecast is plain arithmetic on past records, so every number on screen can be recomputed by hand. No machine learning is used.

Step 1: Predicted breakfast guests (F1). The attendance rate is the average, over the same weekday in the last 4 weeks, of breakfast guests divided by expected guests.

> **Predicted guests = Expected guests tomorrow × Average attendance rate (same weekday, last 4 weeks)**

Step 2: Suggested quantity per dish (F2). Portion per guest comes from past consumption, excluding stock-out days, because on those days consumption hides real demand.

> **Portion per guest = Past quantity consumed ÷ Past breakfast guests**

> **Suggested quantity = Predicted guests × Portion per guest × (1 + buffer)**

Results are rounded to each dish's unit: whole pieces for idli, dosa and poori; 0.5 kg for pongal; 0.5 litre for sambar. The default buffer is 10%.

Worked example (illustrative): 100 expected guests × 84% attendance = 84 guests. At 2 idlis per guest with a 10% buffer: 84 × 2 × 1.1 ≈ 185 idlis.

Data each day (synthetic for the demo): date, weekday, expected guests, breakfast guests, and for each dish: quantity prepared, consumed, discarded, and stock-out (yes/no). Field name is `expected_guests` everywhere, meaning guests, not rooms.

4. Success metrics

All three metrics are shown live in the demo. They prove the product works on synthetic data; they are not claims of real-world accuracy or waste reduction.

Metric	Target	How it is shown in the demo	MRD need
Time to a prep plan	Under 30 seconds from 1 input	Chef types tomorrow's expected guests; we time it live until all 5 dish quantities appear	N1, N2
Back-test accuracy	Predicted guests within ±10% of actual on each of the last 7 synthetic days	The screen forecasts each of those 7 days from the data before it and shows predicted vs actual	N1, N4
Hand-verifiable	A judge recomputes 1 dish quantity by hand and matches the screen exactly	The explanation panel shows attendance rate, portion per guest and buffer used	N4

5. User flows

All three flows run on one dashboard screen, in the order the chef needs them at 9 PM.

F1: Guest forecast

1. Chef opens BreakfastWise; the banner reads "SYNTHETIC DEMO DATA — NOT REAL HOTEL DATA".
2. Chef picks tomorrow's date; the weekday is shown.
3. Chef enters tomorrow's expected guests (for example, 110).
4. The screen shows predicted breakfast guests first, in large type.
5. Below it, the explanation panel shows the expected guests, the 4-week same-weekday attendance rate, and the multiplication.
6. If the input is empty, zero or not a number, the screen shows an error and no forecast.

F2: Suggested prep quantities

1. After step 4 of F1, a table lists the 5 dishes.
2. Each row shows portion per guest, quantity before buffer, and suggested quantity with the 10% buffer, in the dish's unit.
3. A note under the table states that stock-out days were excluded from portions.
4. Chef reads the quantities and plans tomorrow's preparation.

F3: Waste and stock-out history

1. Chef scrolls to the history section.
2. A table shows the last 7 days per dish: prepared, consumed, discarded.
3. Rows where the dish ran out are flagged as stock-outs.
4. A total of discarded quantity per dish is shown for the 7 days.

6. Out of scope

These items are deliberately not built in this hackathon and do not appear in any feature or story.

Out of scope	Reason
Machine-learning forecast models	Needs real, validated hotel data and more time; simple averages are explainable and testable in 1.5 days
Hotel management system (PMS) integration	We have not verified which systems our target hotels use or what data they expose

Out of scope	Reason
Inventory ordering and supplier purchasing	A separate problem from deciding how much to cook; adds scope without serving N1–N3
Multiple hotels or properties	Our segment is single independent hotels; one property proves the flow
Real guest or hotel data	Privacy and availability; the demo uses labelled synthetic data only

Cut first if we fall behind: F5 forecast-vs-actual comparison, then F4 adjustable buffer. Neither breaks the chef's 9 PM flow; the Must-Haves still demo end to end.